DAY 07 OF 16 · 3D & 6DOF STREAMING

Organising space: trees & LoD

How to find neighbours among millions of points in microseconds, and how the same idea gives us Levels of Detail for streaming.

Why structure? | k-d tree | Octree | LoD pyramid | Streaming coarse → fine

Book: Ch. 4, 17 · Code: day07_spatial_structures_lod/

THE BIG IDEA

Most of space is empty. Good data structures spend nothing on emptiness.

At level 10 of our octree, only 0.001% of the possible cells hold any point.

BOOK CH. 4

k-d tree: split space in half, again and again

How

Split the points by the median along x, then y, then z, recursively.

01

Gives

Nearest-neighbour and radius queries in about log N steps instead of N.

02

Lab result

200 queries on 283k points: 449 ms brute force vs 1.3 ms with a k-d tree (335× faster).

03

BOOK CH. 4 & 17

Octree: split a cube into 8, again and again

How

Each node is a cube; an occupied cube gets up to 8 children (octants).

01

Gives

A regular hierarchy: level L has cells of size $S / 2^L$.

02

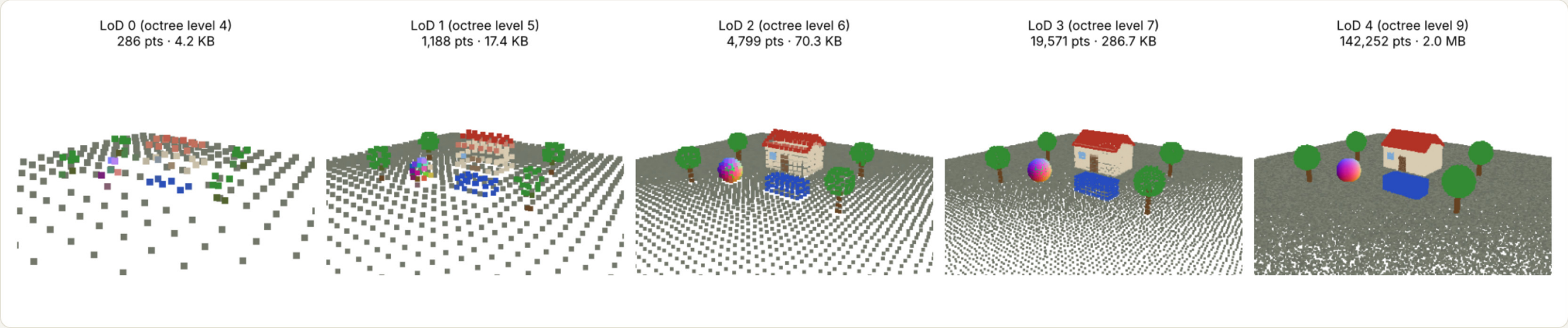
Why for streaming

Each level is a complete, coarser copy of the scene. Download levels in order = coarse to fine.

03

LAB PREVIEW

An LoD pyramid from octree levels



LoD 0: 286 points, 4 KB. LoD 4: 142k points, 2 MB. Each level keeps one point per occupied cell.

LOOKING AHEAD

The octree is a streaming plan

Send coarse first

A preview arrives in milliseconds.

01

Refine where it matters

Only children of nodes the viewer sees and is close to.

02

Standards

Potree, Cesium 3D Tiles and MPEG G-PCC all build on this hierarchy.

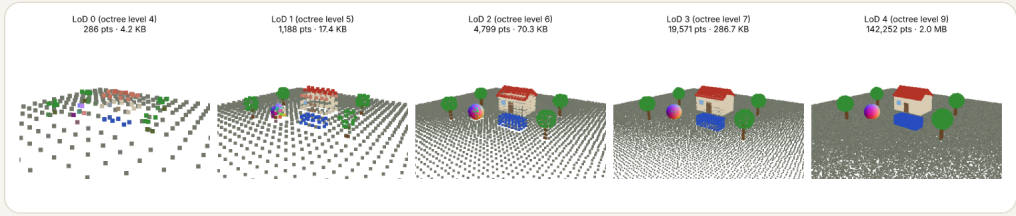
03

HANDS-ON LAB · COPY, RUN, LOOK

Time a k-d tree, build an octree, make LoDs

RUN IT

```
cd day07_spatial_structures_lod
python kdtree_octree_lod.py
```



→ Compare brute-force and k-d tree timings.

→ Read the occupancy table per octree level.

→ Open lod0.ply … lod4.ply in a viewer.

Three things to remember

1. k-d trees make neighbour search ~log N: the engine under most point algorithms.
2. Octrees cut space into 8 recursively and skip empty space.
3. Each octree level is a Level of Detail: the backbone of 3D streaming.

BEFORE NEXT SESSION

Finish mini-project P1 (due Day 8). Start Lab L4. Next: cleaning noisy scans and aligning them.